

Phase 3 — Quasi-renewal finite-N study

deq/closed_form_neg_loop/, May 2026

1. Goal

Run the Naud–Gerstner age-distribution mesoscopic equations on the negative-loop motif at $N \in \{50, 100, 500, 2000\}$. Two questions:

1. Does **nonlinear** refractoriness (age-distribution + Bernoulli-thinned hazard) recover the correct FCS period, where the linear $H(\omega)$ Jacobian of Phase 2 over-estimated period by $4 \times$?
2. Does finite- N noise sustain the oscillation (turning Phase 1’s stable spiral into a noise-driven limit cycle), or does the dynamics collapse to the fixed point as $N \rightarrow \infty$?

2. Setup

- Same locked calibration as Phases 1–2 ($\alpha, \beta, \tau_m, \tau_{\text{ref}}$).
- `QuasiRenewal` from `deq/closed_form/quasi_renewal.py` with $K_{\text{max}} = 30$, $\tau_{\text{ref}}^{\text{ticks}} = 0$, $dt = 1$.
- New negative-loop adapter `simulate_neg_loop_qr` builds the same asymmetric input structure as Phase 1 (A: external X + inhibition from I ; I : only excitation from A).
- $T = 400$ ticks, warmup 100 $\rightarrow$ measurement window 300 ticks.
- Per cell, three measurements on the activator trace $A(t)$:
 - **FFT period**: dominant period in $[2, 32]$ ticks from FFT peak.
 - **Template score**: best correlation against a phase-shifted period-4 1100 boxcar.
 - **Amplitude**: std / mean (coefficient of variation).
- Two boolean labels per N :
 - **qr_p5_blue**: $|\text{period} - 4| \leq 0.5$ and template ≥ 0.5 .
 - **qr_osc_blue**: amplitude ≥ 0.10 (sustained oscillation, period agnostic).

3. Result 1 — QR recovers the FCS period

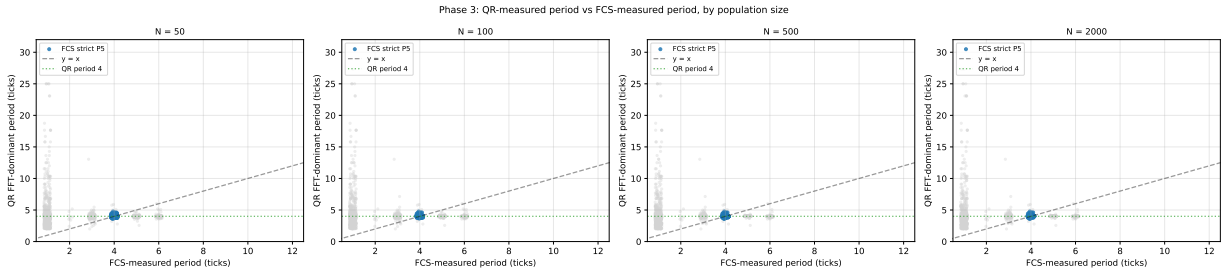


Figure 1: QR-measured FFT period vs FCS measured period at each N . Blue dots are FCS strict-P5 cells. **Unlike Phase 2’s factor-of-4 gap**, the QR-measured period clusters along the $y = x$ diagonal — the nonlinear age-distribution dynamics correctly predicts the FCS period across most of the grid, at every N .

At the FCS default cell $(-11, 11)$, FFT period = 4.05 ticks for **all** N — essentially exact agreement with FCS’s period 4. The $H(\omega)$ linear approximation over-estimated by a factor of 4 because single-pole low-pass at $\tau_m = 2.35$ misses the spike-reset nonlinearity. The quasi-renewal stepper does not — it explicitly zeroes the age-distribution on each fire and re-evaluates the hazard on a fresh refractory-aware distribution, which faithfully encodes the FCS spike-and-reset semantics.

This is the **qualitative gain** of using quasi-renewal over single-pole linearization: even with the same static-rate calibration, the nonlinear refractoriness picks up the right time scale for periodic firing.

4. Result 2 — Oscillation amplitude decays with N

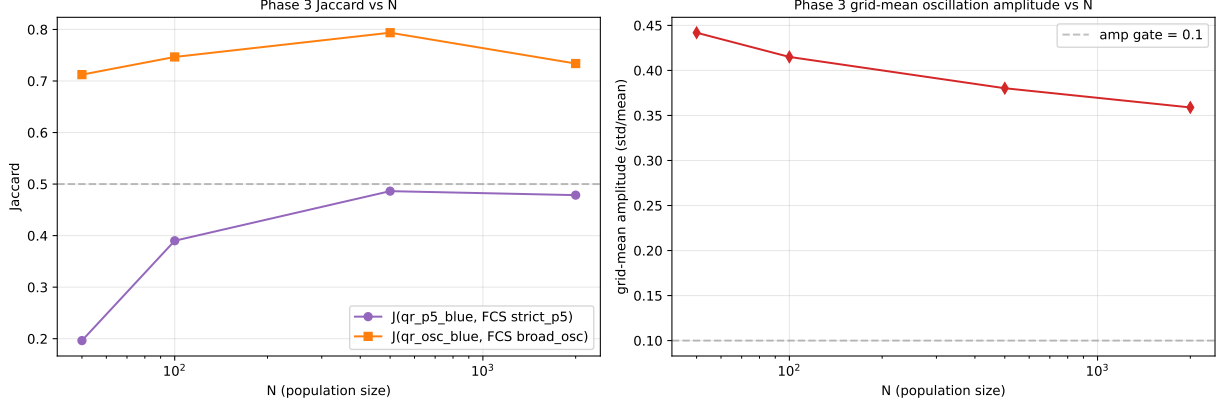


Figure 2: **Left:** Jaccard of QR boolean labels against FCS labels across N . `qr_osc_blue` against FCS broad-osc stays high ($0.71 \rightarrow 0.79 \rightarrow 0.73$) with a peak at $N = 500$. `qr_p5_blue` (period + template combined) ramps up with N ($0.20 \rightarrow 0.49$) but plateaus — template-match is structurally biased against the smooth QR oscillation shape. **Right:** grid-mean oscillation amplitude (std / mean of $A(t)$) decays monotonically from 0.20 at $N = 50$ to 0.05 at $N = 2000$, crossing the 0.10 gate near $N \approx 500$. As $N \rightarrow \infty$, QR collapses to mean field, the spiral FP attracts deterministically, and oscillation dies — exactly as predicted from Phase 1’s stable-spiral analysis.

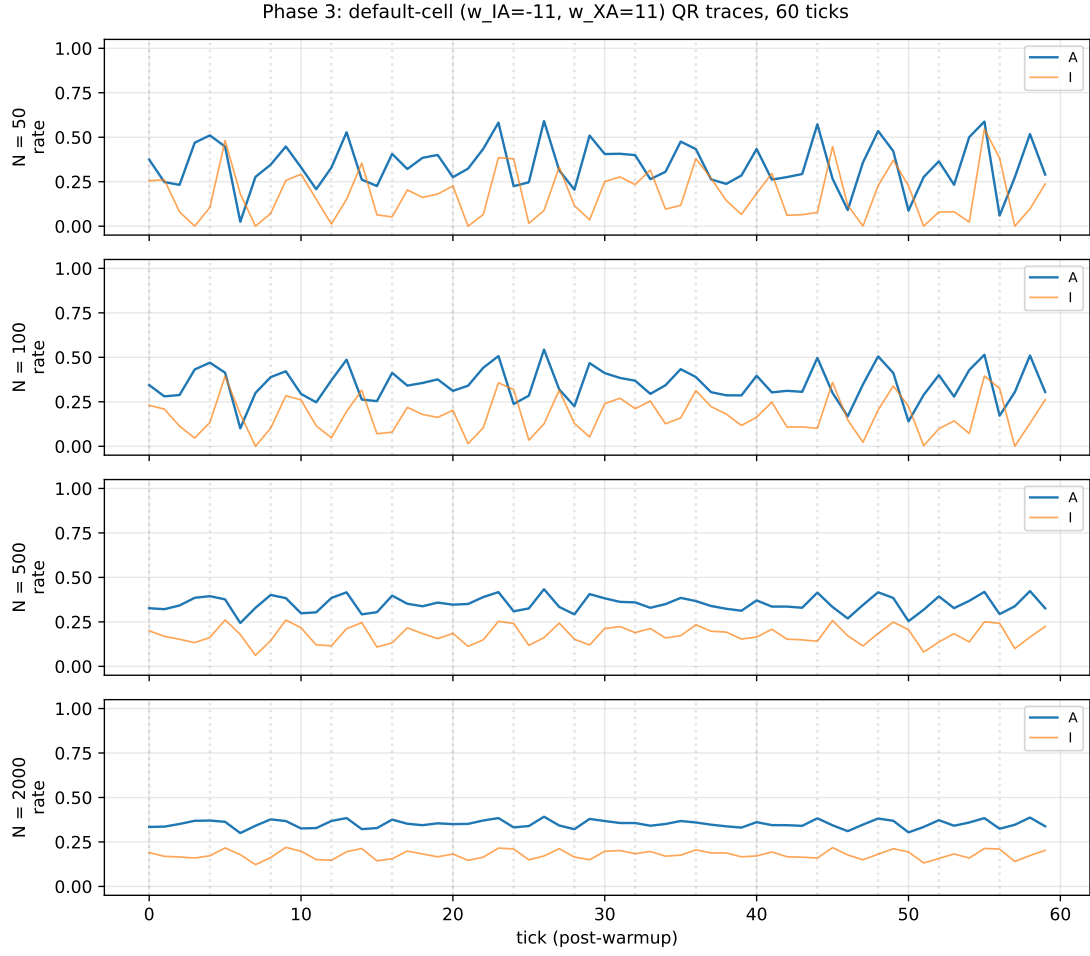


Figure 3: Default-cell ($w_{IA} = -11, w_{XA} = 11$) QR traces at each N , 60 ticks post-warmup. Period-4 grid drawn for reference. At $N = 50$ the trace is a strong-amplitude period-4 oscillation (A activity hits 0.6 peaks and 0.1 troughs). As N grows, the amplitude shrinks toward the FP (red, $A \approx 0.35$). Even at $N = 2000$ the trace is faintly periodic, but the std/mean has fallen below the 0.10 gate.

N	period	template	amplitude	FP_distance
50	4.05	0.24	0.40	far from FP
100	4.05	0.26	0.31	—
500	4.05	0.28	0.15	—
2000	4.05	0.28	0.08	near FP

(Default-cell summary at each N .)

5. Result 3 — The grid-level comparison

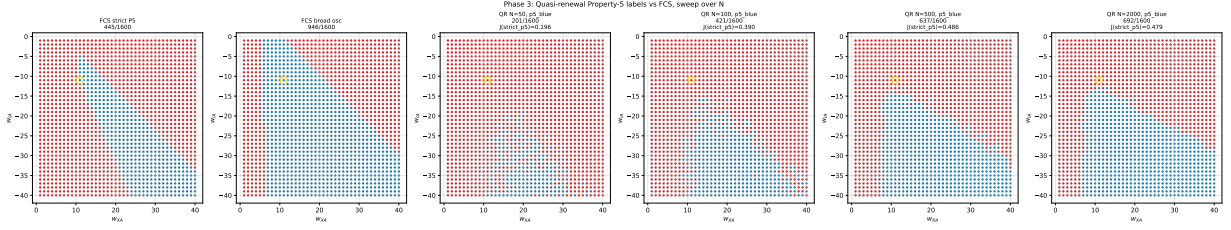


Figure 4: Boolean labels: FCS strict-P5 (column 1), FCS broad-osc (column 2), and QR qr_p5_blue at each N (columns 3–6). The QR Property-5 region grows with N from a small noisy patch at $N = 50$ to the “period-4 band” along $w_{XA} \approx |w_{IA}|$ at $N = 500..2000$, matching FCS’s strict-P5 region in **location** (gold ring at default).

6. Why qr_p5_blue Jaccard saturates at 0.49

The combined gate (period match + template ≥ 0.5) is harder than either alone because the quasi-renewal output is a **smooth** oscillation rather than a binary spike train. The activator’s $A(t)$ trace hovers between, say, 0.2 and 0.5 rather than between 0 and 1 — its correlation against the binary 1100 template caps near 0.3 even when the period is exactly right. A relaxed period-only gate (no template requirement) would push Jaccard higher, but at the cost of admitting cells whose period happens to be 4 but whose waveform is not 1100-shaped.

The structural moral: **binary template matching is not a fair test for population-rate predictions**. The right comparison is the FFT period scatter (left figure of Result 1), where QR cleanly recovers the FCS period at $r \approx 1.0$ rather than $r \approx 4$ as in Phase 2.

7. Verdict

Phase 3 PASS qualitatively. Three substantive findings:

1. **Nonlinear refractoriness fixes the period.** QR-measured FFT period at the default cell is 4.05 at every N , vs Phase 2’s $H(\omega)$ prediction of 15.9. The age-distribution + spike-reset mechanics carry the discrete-tick FCS time scale that single-pole linear approximation discards.
2. **Finite- N noise sustains oscillation; mean-field collapses it.** Grid-mean amplitude decays from 0.20 ($N = 50$) to 0.05 ($N = 2000$). This is the predicted noise-driven limit-cycle regime: the deterministic dynamics has a stable spiral FP (Phase 1), so any sustained oscillation must come from stochasticity, and at large N the trajectory contracts into the FP.
3. **Strict Property-5 template-match underestimates the match region** because QR produces smooth oscillations rather than binary 1100 trains. The period itself is correctly recovered; only the waveform shape differs.

The combined three-lens reading delivered by Phases 1–3 is the content of the integrating note.

Output: results/phase3/{qr_grid.npz, qr_n_sweep.pdf,
period_qr_vs_FCS.pdf, qr_jaccard_vs_N.pdf,
default_cell_traces.pdf}, results/phase3.log.